DIAGNOSTIC BENCHMARK / RESEARCH MANUSCRIPT

Silent Acceptance: A Benchmark of Specification Gates and Extrapolation Failures in Physical Model Assessment

Detection, false alarms, and joint release for prespecified power-law candidates.

Controlled mismatch directions, physics-derived tasks, and an evaluation of extrapolation-related workflow outcomes.

Anonymous - working manuscript, not submitted

Research revision R2 / 2026-10-18

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Silent Acceptance: A Benchmark of Specification Gates and Extrapolation Failures in Physical Model Assessment

Abstract

Specification tests are used to challenge candidate equations, but their value depends on which departures they detect and what errors remain after a model passes them. We benchmark residual-permutation curvature, RESET variants, and nested-quadratic comparisons on prespecified power-law candidates, measuring false alarms, detection, declined evaluations, and extrapolation-related joint release.

On a common pool of 18 non-monomial physics tasks, regressor-power HC3 RESET has the highest mean detection fraction among the four specification tests at each evaluated sample size under the H0-reference 5% calibration target. Its paired task-level advantage over permutation curvature is supported at n=200 and 1000, while constructed odd-symmetric, opposing-curvature, and interaction alternatives reveal distinct diagnostic blind spots. Native false-positive rates also vary with design, sample size, and numerical implementation.

In the 17-task pool with finite extrapolation outcomes, the complete workflow combines $R^2 \geq 0.99$ with four valid, non-rejecting diagnostics. It releases 481 wrong models whose relative error exceeds 10% at the registered upper-end probe: 7.5% of screened dangerous wrong models (481/6,403), or 0.73% of all screened wrong models (481/65,992). The releases are concentrated in two of the seventeen tasks; excluding the threshold-borderline task bonus.15, whose median probe error is 9.2% against the 10% cutoff, the release fraction is 0% at n=200 and

1000. The release fraction among all dangerous eligible fits decreases from 2.94% at n=50 to 0.21% at n=1000. Thus the gates remove many dangerous fits while leaving a measurable residual failure channel whose size depends on task composition and threshold placement.

The benchmark connects diagnostic performance to complete-workflow outcomes and identifies which mismatch directions warrant additional checks. Its results apply to the stated candidate family, task pools, and decision rules; independent joint-policy calibration and evaluation over a fixed extrapolation domain are the next validation steps.

Candidate fit	**Fit screen**	**Gate panel**	**Registered probe**
Prespecified power law	$R^2 \geq 0.99$	No rejection or decline	Far-field error >10%?

Figure 1 - Evaluation stages. Candidate fitting and the fit screen define the screened population; valid gate decisions define release; the registered extrapolation probe measures its consequence.

1 Introduction

A candidate physical equation is useful only to the extent that the available evidence supports its intended use. Prediction error describes agreement on sampled inputs; specification tests ask whether selected departures from the assumed form remain detectable. An engineering workflow often combines the two: fit a candidate, require a high fit score, run diagnostics, and retain models that trigger no alarm. The practical question is how much protection that decision rule provides, and which errors it still releases.

The distinction is concrete in a controlled friction example. A power law fitted to Prandtl-Karman data reaches $R^2 = 0.99$, passes cross-validation and an information-criterion comparison, yet incurs about -20% error one decade beyond training. In the corresponding synthetic scenario, the four specification tests studied here detect the departure at power ≥ 0.99. This motivates evaluating the diagnostics alongside fit quality: a fit-only failure can be detected, but other mismatch directions may survive the same checks.

We study this behavior for prespecified power-law candidates. Permutation curvature compresses departure from log-linearity into one quadratic direction; RESET adds powers of fitted values or regressors; nested-quadratic comparisons evaluate a richer rival using cross-validation or AICc. These choices create testable differences in sensitivity. False alarms and numerical stability also matter: an alarm that detects more wrong forms may reject more correct forms, and an unstable inference implementation can distort both sides of the comparison.

The benchmark combines controlled synthetic alternatives, physics equations sampled from SRSD configurations, family-stratified tasks, a family-viability battery, nested-rival comparisons, and observational compatibility probes. Ground truth supplies form labels in the synthetic and physics-derived suites. Repeated datasets measure sampling variability within those tasks; equations provide the units for the task-level comparisons. Decision thresholds, eligible populations, and declined evaluations are reported with each result.

Three findings organize the study. First, diagnostic sensitivity follows the mismatch direction. Opposing curvature and odd-symmetric departures can be weak in a mean-square projection, while interactions expose differences between fitted-value and regressor-power augmentation. On the common 18-task physics pool, SVD-based regressor-power HC3 has the largest mean detection fraction among A-D at all three sample sizes, with positive paired D-A intervals at n=200 and 1000. Second, detecting residual structure and controlling false alarms are separate empirical requirements. Native size differs across designs; a stable numerical path is necessary for a meaningful comparison. Third, complete-workflow release is more informative than gate silence alone. In the 17-task extrapolation pool, 481 dangerous wrong models pass both $R^2 \geq 0.99$ and every valid diagnostic: 7.5% of the screened-dangerous subset. These releases are concentrated in two tasks; excluding the threshold-borderline task bonus.15, the release fraction is 0% at n=200 and 1000, so the pooled 7.5% is not a task-uniform failure rate. The resulting failure channel can be quantified alongside the errors the gates remove.

We make three contributions:

1. **A diagnostic benchmark with explicit decision accounting.** The six core suites provide ground-truth form labels, specified sampling configurations, threshold policies, and typed failure registration. We distinguish simulation precision within tasks from variation across tasks.

2. **A map of detection, false alarms, and mismatch directions.** Controlled alternatives connect method rankings to specific blind spots. Family viability and nested-rival comparisons show how a diagnostic can support subsequent checks, while implementation cross-checks expose a numerical failure that changes the empirical HC3 comparison.

3. **An evaluation of extrapolation-related joint release.** We connect the fit screen, valid gate decisions, and a registered far-field probe on the same fitted models. Explicit conditional denominators separate the dangerous models released among screened candidates from the fraction of all dangerous fits that survive.

Existing symbolic-regression benchmarks evaluate predictive quality, expression complexity, and symbolic or structural recovery [5, 6]. Our experimental object is the behavior of attached specification gates under a fixed candidate family. This focus complements recovery metrics: it measures the evidence supplied by a diagnostic record when the true form is unavailable to the analyst. Validation on expressions chosen by a discovery algorithm is a further step, developed as a research priority in Section 6.

Sections 3 and 4 define the protocol and report the results. Section 5 draws practical lessons about selecting diagnostics and interpreting release. Section 6 identifies the experiments needed for stronger workflow recommendations. Provenance and correction details are collected in Section 7 and Appendix G.

2 Related Work

Specification testing and robust inference. Ramsey's RESET [1] tests powers of fitted values; regressor-power variants examine other directions of misspecification. Heteroskedasticity-consistent covariance estimation [2] and HC3 leverage adjustment [3, 4] address inference under nonconstant error variance. We compare these established tools on log-transformed, physics-derived designs. The numerical audit in Section 5.2 and Appendix G concerns implementation of the HC3-Wald calculation; the statistical correction and its numerical realization must be distinguished.

Model comparison as a diagnostic. Information criteria [12] and cross-validation [13] can compare a candidate with a richer rival. Our S5 comparison freezes a quadratic rival and reports both native and H0-reference-calibrated policies. This establishes a practical alternative to the single-direction curvature statistic. Its comparison with A is interpreted using the measured error budgets in Section 4.7.

Resampling and transformed coordinates. Wild-bootstrap methods [10, 11] provide tools for inference under heteroskedasticity. Residual permutation relies on exchangeability-related conditions. The family-viability experiment examines how these choices behave after linearizing transformations: replacing permutation with a wild null substantially reduces the observed rational-coordinate distortion. The reported calibration evidence is empirical for the tested designs.

Symbolic-regression evaluation. SRBench [5] evaluates predictive accuracy, expression complexity, and symbolic recovery. SRSD [6] revises physically motivated sampling configurations and evaluates structural discrepancy using normalized edit distance. We reuse SRSD configurations to assess diagnostic decisions for prespecified power laws. The measurements concern detection, false alarms, and extrapolation-related release rather than the recovery rate of a search algorithm.

Physics-informed discovery. AI Feynman [7] uses physical units to constrain search; measurement-driven approaches recover governing structure from observations [8]; PySR [9] supplies a symbolic-search framework. Dimensional constraints and specification diagnostics provide different information. A dimensionally valid non-monomial law still misspecifies a power-law candidate. Our benchmark examines that form mismatch after fixing the candidate family.

Position of the benchmark. The study contributes a common experimental record linking selected diagnostic gates to mismatch directions and complete-workflow outcomes. Its comparative scope covers permutation curvature, fitted-value and regressor-power RESET, HC3 inference, and one nested-quadratic rival. Other residual tests, rival families, and post-selection procedures can be evaluated within the same accounting framework. The experiments establish findings for the tested gates and sampling designs; the wider specification testing literature remains the context for interpreting them.

3 Benchmark Design

3.1 Design principles

P1 - Explicit decision policies. Each comparison identifies the statistic, threshold, eligible population, and handling of declined evaluations. Native p-value rules use alpha=.05. The permutation curvature p-value uses B=999 residual permutations; RESET uses an F reference. S5 compares native selection thresholds with H0-reference-calibrated thresholds. These policies are labeled separately in the results.

P2 - A common reference target for headline power. Table 3 calibrates each method to the empirical 5% quantile of its finite H0 p-values within the same sample-size stratum. Section 4.2 states the inclusive quantile convention and the resulting scope of the comparison. Native-size results, the mixed policies of S5, and joint panels retain their measured error rates. Independent evaluation of the final joint policy is a validation priority in Section 6.

P3 - Structure-defined labels. H0 denotes a strict monomial whose mean is exactly linear in log-log coordinates; H1 denotes a non-monomial ground-truth structure. The symbolic registry supplies these labels rather than diagnostic outcomes. S3 family labels and S4 truth families likewise derive from their generators. Operational eligibility additionally requires valid data and the recorded consistency checks.

P4 - Explicit failure registration. Invalid evaluations retain a typed reason: zero-residual degeneracy, rank deficiency, insufficient degrees of freedom, leverage or covariance degeneracy, statistic mismatch, generation failure, or unavailable observational data. Each table states its evaluated denominator. A declined diagnostic cannot produce a rejection or contribute to joint release. Zero-noise cells exercise the degeneracy path.

P5 - Source sampling configurations. S2/S3 reuse the source registry's per-variable sampling objects. Tasks incompatible with the positive-input/output requirement are marked not applicable. The observational suite uses World Bank Open Data [14], with its indicators and cached snapshots recorded in the artifact.

3.2 Gates under test

All gates operate in log-log space on positive data: let ly = log y and X = [1, lx_1, …, lx_k] be the design of the candidate power-law model.

A. Permutation curvature (form_check). Adds a single mean-centered quadratic column q = $\text{mean}_j((lx_j - \bar{lx}_j)^2)$ and takes the added coefficient c_2 as statistic; size evidence comes from B = 999 residual permutations (vectorized through the pseudo-inverse row), with a statistic-mismatch sentinel comparing wrapper and native values on every call. Validity note: permuting *estimated* residuals assumes they are exchangeable, which OLS residuals are not exactly - the permutation null here is an approximation whose quality is an empirical question, not a finite-sample exactness theorem. The benchmark measures this construction's empirical size on the specified H0 pools, including deviations from .05 (Tables 1-2). Log-transformed multiplicative noise and favorable designs may matter, and §6 lists the stress tests (leverage-adjusted residuals, wild and block nulls, cluster dependence, fixed-design variants) that would bound where the approximation breaks.

B. RESET fitted, classical F. Adds $\hat{y}^2$ and $\hat{y}^3$; classical F test of joint significance.

C. RESET fitted, HC3-Wald. Same augmented design; HC3 sandwich Wald statistic on the added coefficients, referred to F(q, n − k).

D. RESET exog, HC3-Wald. Built regressors are powers of each log-regressor; HC3-Wald as in C.

E. CV-quad (S5). The "obvious alternative": rival = full quadratic expansion of the log features (linear terms, per-feature squares, pairwise cross-terms; minimum-norm least squares absorbs rank deficiency). Statistic D = RMSE_cv(power) − RMSE_cv(quad) over 5 deterministic folds; escalate if D > δ. Naive δ = 0; calibrated δ* chosen on the H0 pool to FPR = 5%.

F. AICc-quad (S5). Same rival; escalate if AICc(power) − AICc(quad) > m, k = column rank. Naive m = 0; calibrated m* as in E.

Configurations were frozen before all main runs - powers (2, 3); B = 999; α = 0.05; the rival family and criteria of E/F; the grids below - and are identical across suites. The same harness invokes all gates on every dataset, so comparisons share data, seeds, and failure handling exactly.

3.3 Task suites S1-S3

S1 - Synthetic control (19 scenarios × 500 seeds). Three monomial H0 scenarios (univariate; two-variable; two-variable with ρ ≈ 0.9 correlated design). Eleven structural H1 scenarios: quadratic curvature (c = 0.10); cubic (standardized odd term); multiplicative interaction (log-space cross term); threshold (exponent 0.4 → 1.0 at the domain midpoint); exponential law; logarithmic law; rational (saturating) law; omitted variable; dilution (one of two variables curved); opposing curvature (+0.12, −0.12 on an exchangeable design, so the mean-square column's population coefficient is exactly zero); collinear-quadratic (ρ ≈ 0.95 design with curvature). Five noise-layer H1 scenarios have strict-monomial mean structure but non-Gaussian noise - Student-t(3); SD growing with the log prediction (0.3× to 1.7×); 2% point outliers (+8σ); 6% group outliers (+5σ); AR(1) ρ = 0.6 - rejections there are false alarms by construction and are never pooled into structural power. Grid: n ∈ {50, 200, 1000}; relative Gaussian noise σ ∈ {0, 0.5%, 1%, 3%, 5%} in log space; 500 seeds per cell via a seed rule embedding the grid position.

S2 - SRSD-Feynman tasks (32 originally fitted tasks, 500 seeds). The original positive-input screen selected 13 monomial H0 and 19 non-monomial H1 tasks from the 120-equation registry. Each fit has a 2,000-sample same-distribution test set. The extrapolation code constructs 12 log-spaced probe points along the first feature: six below the observed training minimum (down to minimum/3) and six above the observed maximum (up to 3 times maximum); all other log-features are held at their training medians. The cached far-field metric used in this paper is the relative error at the last, upper-end point only, against the noiseless closed-form truth. It is not a one-decade probe or the maximum error over the probe grid. Its coordinate depends on the observed training maximum and can change with n and seed. The generated fitted-run count is 240,000; downstream eligibility excludes i.30.5 after its sentinel failure and excludes i.26.2 only when a finite extrapolation outcome is required.

S3 - Family-stratified Feynman equations (30 equations × 30 seeds). Structural family classification from symbolic form: exact power law (12 H0); fractional power (1 H0); exponential (3 H1); rational (6 H1); trigonometric (5 H1); plus not-applicable members with explicit reasons. Each H1 equation runs over relative noise {0.5%, 1%, 3%, 5%} × {homoskedastic, heteroskedastic} × sampling-range scale {official, ×0.5, ×2}; H0 equations run the noise grid at official range.

3.4 Suite S4 - the family-viability battery

The battery evaluates four families through their linearizing coordinates: power (log x, log y), exponential (x, log y), logarithmic (log x, y), and rational (1/x, 1/y). Each member applies A's quadratic construction in its own coordinates and returns a viable set V={F: p_F ≥ alpha}. Columns are standardized before squaring. The formal run uses a wild-bootstrap null with Rademacher signs and B=999. Section 4.6 compares this choice with residual permutation.

The set-level outcomes are false exclusion of a true family, inclusion of a wrong family, and escalation when no family survives. Escalation is correct when the truth is outside all four families. The surviving set supplies candidates for further comparison rather than a unique selected model.

The grid has n=200, relative noise from 0.5% to 5%, 500 seeds per cell, and 188 cells (94,000 datasets). It includes the 19 S1 scenarios with generator-defined truth-family labels and 14 SRSD H1 equations under

homoskedastic and heteroskedastic noise.

3.5 Suite S5 - the gate comparison

S5 compares permutation curvature with the full-quadratic rivals E (cross-validation) and F (AICc), as specified in gate_compare.py. Its pools comprise the 19 S1 scenarios and the 32 originally fitted S2 tasks, with n in {50, 200}, rel=.03, and 200 seeds per cell. A, E, and F evaluate identical datasets.

A uses its native permutation rule. E/F are evaluated at both native selection thresholds and per-pool, per-n H0-reference thresholds. Tables 11-12 report measured size and detection under these policies. Because the resulting budgets differ, the comparison is interpreted using the reported H0 rates.

Reported outcomes include structural-H1 detection, noise-layer false alarms, task-level detail, and computational cost. The S7 reanalysis uses the same records to compare A alone with the A/E/F panel and reports the accompanying change in joint H0 rejection.

3.6 Suite S6 - real data (World Bank open macro data)

Synthetic and equation-derived suites have ground truth by construction; the deployment question is whether the synthetic failure-mode map predicts anything on observational data, where no truth exists. S6 is deliberately small, prespecified, and divergence-oriented: six macroeconomic relations (target, features) × four years (2019, 2021, 2022, 2023), cross-country cross-sections of 184-199 sovereign economies (World Bank aggregates excluded), from eight indicators pulled from the World Bank API with raw responses cached for reproducibility - life expectancy ~ GDP per capita (PPP); infant mortality ~ GDP per capita; GDP ~ population; CO_2 per capita ~ GDP per capita; internet penetration ~ GDP per capita; and life expectancy ~ GDP per capita + health expenditure per capita (two-feature). Gates A-D run at 5 permutation seeds per cell. These seeds are computational replicates of one p-value estimator on one fixed dataset - not five independent samples - so they are used only to report Monte Carlo stability (the empirical p has resolution 1/1000 at B = 999; across the 24 S6 cells the median p spread is .009 and the max .094), never as a sample-size or stability claim; a prespecified probe adds three agricultural and energy relations over three years, including a built-in control-contrast pair - cereal yield ~ fertilizer (one feature) versus cereal yield ~ fertilizer + energy use per capita (two features) - chosen to test, on live data, whether the dilution mechanism of S1 (a second correlated curved feature halving the mean-square column's signal) operates outside the laboratory. No calibration or power claim is made on S6; its contract is external validity of *mechanisms*, not of rates. An engineering companion bench (Appendix E) extends the same contract to physical-engineering data: nine open sources, 91 datasets, 150 prespecified relations, run as a single-realization smoke pass.

3.7 Suite S7 - joint-silence reanalysis of existing results

S7 evaluates panel non-rejection before imposing the fit screen. The A-D arm joins S2 records to SVD C/D outputs one-to-one on (task, n, rel, seed), requires positive noise and a valid statistic sentinel, and contains 18 H1 tasks and 13 H0 tasks. At each n in {50, 200, 1000}, a task contributes four noise levels times 500 seeds. Native alpha=.05 and per-gate H0-reference 5% thresholds define separate policies. Joint silence requires finite p-values for every gate and no rejection.

The A/E/F arm uses the S5 SRSD pool at rel=.03: 19 H1 tasks and 13 H0 tasks, with 200 seeds each at n=50 and 200. A retains its native rule; E/F use their individual H0-reference thresholds. A alone and A/E/F are compared on the same records, with the change in joint H0 rejection reported alongside detection.

The two arms have different task pools, noise levels, seeds, and gate sets. They estimate their own panel behavior; only the paired within-arm comparison measures the effect of adding E/F. Both arms reuse existing fitted records. Reference thresholds and the reported H0 rates use the same pools, so the evidence describes reference-pool calibration. audit_s7.py records filters and exact counts.

Section 4.10 additionally requires finite fit-quality and far-field outputs, yielding 17 H1 tasks. This defines the joint-release pool; S7's gate-only analysis uses all 18 eligible H1 tasks.

4 Results

Tables 1-2 report size; Tables 3-4 detection; Table 5 the family map; Table 6 noise-layer behavior; Table 7 extrapolation; Tables 8-10 the battery; Tables 11-12 the model comparison; Tables 13-14 observational examples; Table 15 joint release; and Table 16 joint silence. Each table defines its event, denominator, task pool, and failure policy. Computational repeats are not independent equations. S7 reuses S2/S5 records rather than adding new fits. Sections 4.2, 4.10, and 4.11 apply the eligibility rules in Section 3 and the accompanying record-level audit scripts.

4.1 Size: false-positive rates on true monomials

Table 1 - S1 H0 false-positive rates at p < .05. Three scenarios, four positive noise levels, and 500 seeds give 6,000 valid evaluations per n. Entries are rates with fit-level Wilson 95% intervals.

Method	n=50	n=200	n=1000
A	.0583 [.0527-.0646]	.0457 [.0407-.0512]	.0513 [.0460-.0572]
B	.0557 [.0501-.0618]	.0518 [.0465-.0577]	.0472 [.0421-.0528]
C (SVD)	.0602 [.0544-.0665]	.0568 [.0513-.0630]	.0478 [.0427-.0535]
D (SVD)	.0675 [.0614-.0741]	.0525 [.0471-.0584]	.0502 [.0449-.0560]

The evaluated-H0 denominator excludes 1,500 zero-noise records per n, each registered as degenerate_zero_residual. Such a record has no valid test decision and is not a successful non-rejection.

Table 2 - S2 H0 false-positive rates at p < .05. Thirteen equations, four positive noise levels, and 500 seeds give 26,000 valid evaluations per n; C/D use SVD.

Method	n=50	n=200	n=1000
A	.0489 [.0463-.0516]	.0535 [.0508-.0563]	.0517 [.0491-.0545]
B	.0499 [.0473-.0526]	.0497 [.0471-.0524]	.0527 [.0500-.0555]
C (SVD)	.0565 [.0538-.0594]	.0531 [.0505-.0559]	.0549 [.0522-.0577]
D (SVD)	.0604 [.0576-.0633]	.0564 [.0536-.0593]	.0583 [.0555-.0612]

Both tables are generated from existing positive-noise H0 records by audit_size.py. Stored rejection flags agree with strict p < .05; ties at exactly .05 are not rejections. This differs from the inclusive empirical-quantile convention used in Table 3.

On synthetic Gaussian data, Table 1 shows small-sample size distortion: at n=50 all four fit-level Wilson intervals lie above .05, while the pattern is milder at larger n. Calibration is therefore empirical and imperfect, not exact. On the physics-derived designs (Table 2), A and B are near .05, while C and D have a modest upward size distortion: all six displayed C/D fit-level Wilson intervals exclude .05. The SVD calculation underlying these values was cross-checked against a separate implementation (Section 5.2).

Generalization over physical equations requires an equation-level uncertainty analysis, distinct from a fit-level interval on this fixed task pool. At n=200, resampling the 13 H0 equations gives C=.053 with a 95% interval [.050, .056] and D=.056 with [.055, .058]; the corresponding leave-one-equation-out ranges are [.052, .054] and [.056, .057]. The C lower bound rounds to .050 but is slightly above .05 before rounding. Appendix C uses this same SVD result and n. These intervals describe the observed task pool, not all possible physical laws.

A separate fitted-HC3 implementation check found 8/120 rejections (.067, Wilson [.034, .126]) on two tasks. This small-sample check addresses numerical agreement; Table 2 and the equation-level analysis supply the size estimates for the S2 pool. Appendix G records the original-path comparison and its interpretation.

4.2 Power at a common reference-pool calibration target

Table 3 uses the same 18 sentinel-valid H1 tasks for every method; feynman-i.30.5 is excluded. Each n has 36,000 records: four positive noise levels times 500 seeds per task. Thresholds are the empirical 5% quantiles of each method's finite H0-reference p-values at the same n. Table 3 rejects when a valid p-value is less than or equal to that quantile; native decisions instead use strict p < .05. Non-finite p-values and typed declines are never rejections or certificates. Quantile discretization and boundary ties mean that this is a calibration target, not a claim of exactly equal rejection counts or independent calibration.

Table 3 - S2 rejection fraction on the common 18-H1-task pool at the H0-reference 5% target; C/D use SVD. Declines remain in the denominator and are not counted as rejections.

Method	n=50	n=200	n=1000
A	.386	.486	.570
B	.187	.254	.383
C (SVD)	.128	.221	.383
D (SVD)	**.466**	**.657**	**.718**

For B and C, respectively, 2, 56, and 591 p-values are non-finite out of 36,000 at n=50, 200, and 1000. These records cannot contribute a rejection or a silent certificate. The displayed fractions are unconditional detection yields on the specified records, not power conditional on successful evaluation.

D has the largest mean rejection fraction among A-D in this pool. The paired equation-level bootstrap compares D minus A on these same 18 tasks (10,000 resamples): +.080 [-.083, +.256] at n=50, +.170 [+.042, +.329] at n=200, and +.148 [+.022, +.302] at n=1000. The first interval crosses zero; the latter two favor D in this task pool. D exceeds A on 10/18, 13/18, and 8/18 tasks, with 0, 1, and 5 ties, respectively. A larger pooled mean is therefore not uniform task-wise dominance.

Table 3, its paired effects, and Figures D.2-D.3 use the same eligibility rule. S5 evaluates the nested-rival gates on a separate pool; a paired D-versus-F comparison is a validation priority. The constructed alternatives below illustrate the method-specific differences behind the pooled ranking.

Table 4 - S1 blind-spot cartography (raw α, n = 200, rel = 1%).

Scenario	A	B	C	D
cubic (standardized odd term)	.40	1.00	1.00	1.00
interaction ($lx_1 \cdot lx_2$ cross term)	.21	1.00	1.00	.10
opposing curvature (+c, −c, exchangeable)	**.02**	.98	.98	1.00
dilution (one curved variable of two)	1.00	.88	.87	1.00

Three mechanisms, each verified by construction: (i) A's statistic is the mean of squared centered columns - under opposing curvature on an exchangeable design that mean has exactly zero population signal; (ii) a quadratic-projection statistic loses power on odd-symmetric deviations; (iii) B's fitted-value powers are a local Taylor surrogate for any smooth alternative - saturating on interaction where A (.21) and D (.10) struggle - and, as S3 shows, the same surrogate property near-blinds B on rational laws.

4.3 Family-stratified results and sensitivity

Table 5 - S3 calibrated power by family (C/D on the verified SVD path).

Family	A	B	C (SVD)	D (SVD)
exponential	.09	.06	.07	.06
rational	**.65**	.16	.13	**.67**
trigonometric	.66	.69	.58	**.86**

B - near-perfect on synthetic quadratics - is nearly blind to rational laws (.16 calibrated): the fitted-value Taylor surrogate absorbs precisely the low-order curvature by which a saturating law departs from a power law. Under the SVD path D is the strongest gate on both rational (.67) and trigonometric (.86) families; the previous draft's near-zero C/D entries on trigonometric were the pinv artifact (thresholds derived from artifact p-mass), and the S3-level correction agrees in direction with the S2-level one of Table 3. Exponentials defeat every gate (≤.09 - the linearization a log transform induces on them is nearly orthogonal to all four augmentation families); A and D share the rational lead; no method dominates all three families. Every method's strength sits on another method's blind spot; Table 5 has no row-major winner. Sensitivity: scaling every sampling domain ×2 moves exactly one family-method cell by more than 15 points (rational × A: .66 → .84, the expected direction); ×0.5 moves none; heteroskedastic replications agree within .02 on every cell. **These sensitivity readings rest on 30 seeds per cell and cannot be separated from sampling noise at that size**: a single cell at p ≈ .66 carries a 95% interval of roughly ±.17, and the standard error of a difference between two independent 30-seed cells is ≈ .11, so a 15-point trigger cannot distinguish movement from variation. Family × method intervals are now attached (`paper/review3_analysis.py` §3.8b). On the within-cell binomial convention the trigonometric D (SVD) cell is .860 [.848, .871] and rational D is .669 [.655, .683]; on the honest unit - the equation - the same cells read .860 [.592, .998] and .669 [.409, .864]. Rational C moves from .132 [.122, .142] to [.053, .277] the same way. The equation-level intervals overlap across families and are wide enough to contain orderings the point estimates contradict, so the family ranking above is directional, not an ordered effect list; what survives the wider intervals is the sign pattern - D ≥ C on rational and trigonometric, every gate low on exponential - not the point ordering.

4.4 Noise-layer robustness

Table 6 - S1 noise-layer H1 (strict-monomial mean structure) rejection rates at α = 0.05 - false alarms attributable to the noise assumption (n = 200, rel = 1%, 500 seeds; C/D on the verified SVD path).

Scenario	A	B	C (SVD)	D (SVD)
Student-t(3) noise	.034 [.021-.054]	.036 [.023-.056]	.030 [.018-.049]	.030 [.018-.049]
variance growing with prediction	.048 [.032-.070]	.058 [.041-.082]	.050 [.034-.073]	.050 [.034-.073]
AR(1) errors, ρ = 0.6	.038 [.024-.059]	.040 [.026-.061]	.042 [.028-.063]	.042 [.028-.063]
2% point outliers (+8σ)	.040 [.026-.061]	.066 [.047-.091]	.036 [.023-.056]	.036 [.023-.056]
6% group outliers (+5σ)	.038 [.024-.059]	.060 [.042-.084]	.054 [.037-.077]	.054 [.037-.077]

Wilson 95% intervals use 500 seeds per cell. C/D are computed from the cached SVD results; paper/review3_analysis.py §3.8c documents the cell-level check.

The recorded rates span .030-.066, with Wilson intervals shown (500 seeds per cell). These selected S1 perturbations do not isolate the causes of the S2 size distortions. Neither noise-related effects nor interactions between noise, design, and inference are ruled out.

4.5 Do alarms predict extrapolation failure?

The S2 extrapolation pool contains 180,000 records: 13 H0 and 17 H1 tasks, positive noise, and all three sample sizes. It excludes sentinel-invalid or missing-outcome tasks. Danger here means the registered upper-end point error exceeds 10%, without an R^2 screen; 36,206 records meet it (base rate .2011), including 37 records from true-form H0 tasks.

Table 7 - Alarm precision and recall for >10% upper-end extrapolation error on the S2 pool, native thresholds; C/D use SVD. Declines produce no alarm and remain in the recall denominator.

Method	precision	recall	declines
A	.4226	.6242	0
B	.6379	.4901	649
C (SVD)	.6189	.4639	649
D (SVD)	.4879	.8844	0

A and B have 10,296 and 5,442 exclusive true-positive alarms, respectively, with 12,304 shared. These counts show different alarm sets, not a claim of independence or chance-level overlap. D detects more of this pool's dangerous outcomes, but the precision/recall table does not compare methods at an equal false-positive budget. Figure D.4 uses the same eligible pool. An alarm and a non-rejection can both be informative under a specified sampling distribution; neither supplies a bound on error throughout an extrapolation domain.

4.6 The battery: from rejection to routing

Table 8 - Probe: permutation vs wild null by family (n = 200, 200 seeds; entries rel = 1% / 3%).

Truth	perm null	wild null
power (size of power member)	.070 / .075	.075 / .060
rational: false escalation (V = Ø)	**.34 / .31**	.045 / .090
rational: containment	.66 / .70	**.96 / .91**
exponential: containment	.94 / .95	.92 / .95
logarithmic: containment	.87 / .93	.89 / .95
threshold (no battery family): escalation - *correct outcome*	1.00 / 1.00	1.00 / 1.00

With the permutation null, the battery's own rational member rejects its own true family a third of the time - nothing about rational laws is at fault; the null is exchangeability-invalid in (1/x, 1/y) coordinates. The wild bootstrap repairs exactly that and nothing else.

Set-level errors and multiplicity. The battery's decision is the viable *set*, so the errors that matter are set-level, and each member runs at $\alpha = .05$ without correction - a union decision whose two error types move oppositely under multiplicity. Two pools, two behaviors. On H0-side cells (truth is a battery member) the reported containment is one minus the exclusion rate of the true family's own member (.0496-.066 across the four truth families; the viable set is never empty *on this pool*), and per-member α does not inflate that rate because exclusion depends on one member only. Inflation appears on the other side: some wrong member survives alongside the truth in 23.3% of power-truth cells (near-neighbour overlap: exponential/logarithmic members retain power data) and 92% of logarithmic-truth cells (rational retains log data) - reported as near-neighbour overlap rather than suppressed in the locally recorded tables. On the structural-H1 pool (72,000 cells whose truth is no battery member by design) the viable set is empty - $V = \emptyset$, correct escalation - at the rates of Table 10's first column: universally for quadratic/threshold/omitted-variable (1.00), rarely for the interaction/opposing/collinear structures (.001-.05) whose mismatch directions the member families do not span. Routing there means *which wrong family is modal* (Table 10, second column); the battery is a candidate filtering stage whose surviving set the discovery layer must still rank, not a final selection. Downstream selection error after the battery is out of scope here and unmeasured.

Table 9 - Battery on synthetic truths (500 seeds/cell, wild null).

Truth family	Containment (truth $\in$ V)	False escalation (V = Ø)
power - size, 8 scenario groups incl. all noise layers	-	.0496 [.046-.053] pooled; per-scenario .043-.054
exponential	.95	.05
logarithmic	.94	.005
rational	.93	.066 pooled ($\leq$ .082 worst rel)

Table 10 - Battery on out-of-family structures (escalation to V = Ø is the correct outcome; column 1 is the V = Ø rate).

Scenario	Correct escalation	Modal false V
quadratic / threshold / omitted variable	1.00 / 1.00 / 1.00	-
dilution	.92	rational
cubic (odd term)	.24	power (.68)
interaction	.05	power (.95)
opposing curvature	**.00**	power+rational (.84)
collinear-quadratic	.03	rational (.61)

Table 10 is Table 4 re-expressed as routing errors: the power member inherits A's documented blind spots, so those structures stay in V under the power label instead of escalating. On the SRSD side the routing is reported descriptively (median |V| 1-2 for 11 of 14 equations; i.30.5 → {exponential} in 78%; ii.38.14 escalates on 67%), because surface-form family labels dissolve under examination - the battery correctly rejects Planck-type "exponentials" that no linearizable family owns.

4.7 The gate comparison: why not just compare against a nested rival?

Table 11 - S5 size on H0 pools (200 seeds/cell; naive and calibrated thresholds for E/F).

H0 pool	A	E naive	E cal.	F naive	F cal.
synth n=50 (600 fits)	.080	.212	.050	.132	.050
synth n=200 (600)	.040	.183	.050	.135	.050
SRSD n=50 (2,600)	.051	.167	.050	.086	.050
SRSD n=200 (2,600)	.038	.152	.050	.123	.050

Table 12 - S5 structural-H1 detection (A at its native threshold; E/F at H0-pool-calibrated thresholds; mean over tasks).

	synth n=50	synth n=200	SRSD n=50	SRSD n=200
A (permutation curvature)	.774	.782	.348	.493
E (CV-quad)	.929	.934	.462	.628
F (AICc-quad)	**.930**	**.940**	**.525**	**.669**

OR(A, E) at each gate's own 5% threshold - a different, larger joint error budget (.065-.102 across the four benches), reported separately: .937/.941/.547/.675. It is not comparable to the rows above and is excluded from this table.

The error budgets are not exactly matched in Table 12: A retains its native threshold, whereas E/F are tuned on the H0 reference pool. In particular, A's synth n=50 size is .080, and its SRSD n=200 size is below .05. Interval overlap with a target is not proof of an identical budget. Table 12 reports measured detection under these policies, not a uniformly matched-FPR ranking.

The quadratic rivals include per-feature squares and cross-terms, covering directions absent from A's single mean-square column. On the recorded S5 pool, F has the largest mean detection in each displayed column. This is a scoped finding for one rival family, not a universal default recommendation.

Naive E/F thresholds have elevated measured H0 rejection rates (Table 11). Reference-pool calibration reduces them to the reported targets; A requires no separate reference pool to construct its resampling p-value, but that does not make A size-valid on every pool. Its .080 small-sample result is one counterexample.

The earlier SRSD leave-one-H0-equation-out analysis reports held-out sizes .053/.050 for E at n=50/200 and .051/.052 for F, with per-equation spread [.010, .095]. This provides a limited transfer check across the 13 available H0 tasks, not calibration on all new equations or a replacement for an independent

deployment test set.

For OR(A,E), using the individual thresholds increases the joint H0 rejection budget to .065-.102. At synth n=50, A alone already exceeds .05, so tightening E cannot achieve a 5% union budget while preserving A's rule. The reported joint-budget recalibration at n=200 gives .940/.664 detection on the synthetic/SRSD benches, compared with F's .940/.669. Those specific comparisons do not establish that F is the frontier at every possible error budget. Combining gates and selecting thresholds should be evaluated as a complete decision policy.

4.8 Real data: does the map predict live behavior?

Table 13 - S6 World Bank relations: verdicts (median permutation p over 5 seeds for A; deterministic p for B-D; α = .05).

Relation (target ~ features)	Years all four gates agree	Divergent cells (which gate, which way)
infant mortality ~ GDP pc	4/4 reject	-
CO_2 pc ~ GDP pc	4/4 reject	-
internet penetration ~ GDP pc	4/4 reject ($p < 10^{-16}$)	-
GDP ~ population	4/4 **quiet** (A .71-.91; B .19-.33)	-
life expectancy ~ GDP pc	0/4	A alone rejects 2019 (.035) & 2022 (.008); B alone rejects 2022 (.0002); C/D never
life expectancy ~ GDP pc + health exp pc	2/4 quiet	B rejects 2022 (.0002) while A quiet (.108)

Where signal is strong the gates agree - three relations rejected unanimously every year, one candidate relation unrejected by all four gates - and A's five p-values per cell are Monte Carlo replicates of one estimator whose spread is small (median .009, max .094 at resolution 1/1000; no verdict flips beyond boundary cells). Divergence concentrates where the synthetic map puts weak curvature: the life-expectancy cells flip year to year (signal at the detection boundary), and the two-feature cell shows B rejecting where A stays quiet. We label this pattern *consistent with* the dilution mechanism (§4.2, Table 4) rather than a reproduction of it: on observational data the p-value change after adding a second feature can also reflect feature correlation, omitted-variable changes, sample composition, measurement error, cross-country heterogeneity, or spatial dependence - no ground truth exists to separate these, so the claim is signature-consistency, not mechanism confirmation.

Table 14 - S6 dilution probe: the same target with and without a second correlated curved feature (3 years × 5 seeds; detection = median gate verdict).

Relation	A	B	D
cereal yield ~ fertilizer (1 feature)	rejects 3/3 years (.011-.031)	rejects 3/3 (~.003)	rejects 3/3
cereal yield ~ fertilizer + energy pc (2 features)	**quiet 3/3 years** (.067-.149)	rejects 2/3 (.025-.092)	rejects 3/3 (≤.018)

This matches the S1 DILUTION scenario's signature - adding one correlated curved feature silences A on a relation it was rejecting, while the RESET variants continue to alarm - a pattern *consistent with* the dilution mechanism rather than a demonstrated reproduction of it, since feature correlation, sample composition, and omitted-variable changes move the same p-values in the same direction. No ground truth exists on S6 and none is claimed; the contract of §3.6 is signature consistency in both directions: unanimous cells where the map predicts signal, and map-consistent divergence where it predicts weakness.

A companion engineering bench (Appendix E) pushes the same probe across nine physical-engineering sources (147 run cells): unanimous detection of a documented rating-curve misspecification, mechanism-echo divergences matching the synthetic map (small-sample power, interaction carried by a second feature, all three HC3 failure modes live), and mutual misses on locally approximate laws - all recorded, none adjudicated.

4.9 Runtime

The historical S1 runs recorded per-dataset means of 12.0-25.3 ms for A and 1.0-1.8 ms for B/C/D; the family battery at n=200 recorded 24 ms, and S5 E/F 1.5-3.5 ms. These figures describe the recorded environments and implementations. No inference-path runtime equivalence or production-throughput guarantee is claimed. Figure D.5 is retained as a historical timing panel; the SVD revision requires a separately labeled timing comparison.

4.10 What the full pipeline actually releases

The joint-release analysis links fit quality, native A/B decisions, SVD C/D decisions, and extrapolation errors one-to-one on (task, n, rel, seed). It requires positive noise, a valid statistic-consistency sentinel, and finite R^2 and far-field outcomes. The 17 eligible H1 tasks contribute 34,000 records per n. The i.30.5 statistic-sentinel records and the i.26.2 records without finite far-field outcomes are excluded under these rules. audit_results.py records the join and all filters.

Decision events. Release requires $R^2 \geq 0.99$, four valid gate evaluations, and no rejection at the native alpha=.05 rules. A dangerous wrong model has relative error >10% at the registered upper-end probe (Section 3.3). A declined diagnostic withholds release. The error threshold defines the consequence evaluated here; the probe measures one input rather than an operating-domain maximum.

Table 15 - Joint release on the 17-task S2 H1 pool. Each n has 34,000 records. Count symbols S, Sd, Ld, and Da distinguish the event populations from gate labels A-D. Eligibility excludes i.30.5 (statistic-sentinel failure) and i.26.2 (no finite far-field outcome); Table 16 uses the 18-task H1 pool without the finite-far requirement.

n	Screened S	Dangerous in S: Sd	Released dangerous Ld	All dangerous Da	Ld / S	Ld / Sd	Ld / Da
50	22,060	2,208	350	11,922	1.59%	15.85%	2.94%
200	21,932	2,150	106	12,202	0.48%	4.93%	0.87%
1000	22,000	2,045	25	12,045	0.11%	1.22%	0.21%

S counts screened wrong models; Sd counts dangerous wrong models within S; Ld counts those passing both the screen and all four valid gates; Da counts every dangerous model in the eligible H1 pool. The three ratios answer distinct operational questions: frequency among screened wrong forms, failure to intercept screened-dangerous forms, and survival among all dangerous fits. Pooled across n, S=65,992, Sd=6,403, Ld=481, and Da=36,169. The respective rates are 0.73%, 7.5%, and 1.33%.

The fit screen changes the release population. At n=50, 966 dangerous records have four valid non-rejecting gates; 616 fail the fit screen, leaving 350 complete-workflow releases. At n=200 and 1000, the gate-only and complete-workflow dangerous release numerators coincide at 106 and 25. Across n, the panel declines on 2, 56, and 591 eligible H1 records. This accounting separates rejection, diagnostic withholding, fit screening, and complete release.

Detection benefit and residual failure. Among screened-dangerous wrong models, 481/6,403 (7.51%) are released. Among screened wrong models below the danger threshold, 27,025/59,589 (45.35%) are released. The panel therefore excludes most screened-dangerous records from release while retaining a substantial residual channel. These pooled rates summarize the sampled task composition, and that composition is concentrated. The 481 releases come from two of the seventeen tasks: bonus.15 (242) and bonus.5 (239). Six of the remaining fifteen release no dangerous record at any n while contributing 30,351 dangerous records between them; among these, all 557 dangerous records of ii.38.14 are intercepted. Structural mismatch and failure at the chosen probe are different labels, and their relation varies by equation.

Sensitivity to the danger threshold. At n=50, thresholds of 2%, 10%, and 20% label 34.91%, 10.01%, and 2.76% of screened H1 models dangerous. The release fractions within those subsets are 41.30%, 15.85%, and 6.40%. At n=1000 and the 20% threshold, Sd=0 and Ld/Sd is undefined. The complete 2/5/10/20/50% sweep is in evidence/results.json.

A pooled task-level sweep makes the composition visible. At the 10% threshold, all 481 releases come from bonus.15 (242) and bonus.5 (239). At 12.5%, the count falls to 176: bonus.15 contributes 5 and bonus.5 contributes 171. At 15%, all 111 releases come from bonus.5; at 20%, all 39 do. The released set therefore consists of the two tasks throughout, but which task dominates changes sharply between 10% and 12.5%. These counts are reproduced by threshold_decomposition.py and recorded in evidence/threshold_decomposition.json.

Sample-size pattern. Ld/Da decreases from 2.94% to 0.87% to 0.21%, and Ld/Sd decreases from 15.85% to 4.93% to 1.22%. A fixed-label control assigns each matched (task, rel, seed) triple its n=1000 danger label. Across 34,000 complete triples, Ld/Sd is then 16.77%, 1.87%, and 1.22%, while Ld/Da is 2.93%, 0.31%, and 0.21%. The decreasing pattern survives this label control. Its interpretation still combines fit-screen behavior, diagnostic sensitivity, and changing probe coordinates; training samples are not constructed as nested sets.

Task composition of the released set. The released records change identity with n. At n=50 the 350 releases are 239 from bonus.5 and 111 from bonus.15. At n=200 and n=1000 every release (106 and 25) comes from bonus.15 alone. Excluding that task, Ld/Sd is 11.58%, 0%, and 0%. The fixed-label control holds the danger label but not the task composition: bonus.15 has a median relative far-field error of 9.2%, with the 75th and 90th percentiles at 9.5% and 9.8% against the 10% threshold. Its danger label therefore moves with the threshold while its fitted form and task identity do not change. The decreasing pattern

combines a panel effect with the position of one task's error distribution relative to a fixed cutoff.

Task-level uncertainty. The record-level ratios treat simulation replicates as independent observations. A task bootstrap instead resamples the 17 eligible equations with replacement (B=10,000), carrying each task's S/Sd/Ld/Da and recomputing the pooled ratios. Pooled over n, Ld/Sd is 7.5% with a 95% interval of [0%, 49.9%]; 12% of resamples contain no released dangerous record, and 15 of the 17 tasks contribute Ld=0. Leave-one-task-out values span [3.9%, 20.1%]: omitting bonus.5 raises Ld/Sd to 20.1%, while omitting bonus.15 lowers it to 3.9%, because the remaining task's own release fraction then dominates. At n=200 and 1000 the interval reaches [0%, 75.7%] and [0%, 56.8%], since only bonus.15 contributes releases. This interval measures between-equation heterogeneity, not binomial precision, and the record-level 7.5% describes the sampled task pool rather than a task-uniform rate. Counts and intervals are reproduced by audit_task_intervals.py and recorded in evidence/task_intervals.json.

Information in a clean diagnostic record. The corresponding 180,000-record H0/H1 pool has three states: true H0 (78,000 records; 65,934 releases), dangerous wrong H1 (36,169; 481 releases), and non-dangerous wrong H1 (65,831; 27,025 releases). The empirical release probability is .013299 for dangerous H1 and .845308 for true H0. Their likelihood ratio is .01573 in the dangerous-versus-true direction, or 63.56 in the reverse direction.

Across all three states, 481 of the 93,440 released models are dangerous wrong forms (0.515%). This proportion depends on the pool's class composition. In a different, explicitly hypothetical population containing only true H0 and dangerous H1 with equal priors, Bayes' rule gives a 1.55% dangerous posterior after release. These are separate populations and their conditional risks must remain separate. Applying the decision rule to a new population requires its own validation.

The results support a complete-workflow evaluation: diagnostics carry information and remove dangerous candidates, while a clean record leaves measurable residual uncertainty.

4.11 Joint silence without the fit screen

Panel non-rejection is measured separately from complete release using the S7 arms defined in Section 3.7. audit_s7.py reproduces these counts from the existing records.

Table 16 - A-D joint silence on S2 records. H1 has 18 eligible tasks (36,000 records per n); H0 has 13 tasks (26,000 per n). Calibrated means a 5% H0-reference target per gate.

n	H1 native alpha=.05	H1 per-gate calibrated	H0 per-gate calibrated
50	13,616 / 36,000 (37.8%)	13,888 / 36,000 (38.6%)	22,821 / 26,000 (87.8%)
200	10,717 / 36,000 (29.8%)	10,877 / 36,000 (30.2%)	23,102 / 26,000 (88.9%)
1000	8,355 / 36,000 (23.2%)	8,478 / 36,000 (23.6%)	23,108 / 26,000 (88.9%)

A silent record has finite p-values for all four gates and no rejection. The H0 rates are measured on the reference pool used to set the thresholds. The resulting joint H0 rejection exceeds 5%, illustrating the distinction between individual calibration targets and the error budget of a panel.

Paired A versus A/E/F. On the S5 SRSD pool at n=50, adding E/F reduces H1 silence from 2,479/3,800 (65.2%) to 1,602/3,800 (42.2%). H0 silence decreases from 2,468/2,600 (94.9%) to 2,352/2,600 (90.5%). At n=200, H1 silence decreases from 1,926 to 1,160 out of 3,800, and H0 silence from 2,500 to 2,377 out of 2,600. The paired comparison demonstrates increased detection together with increased false rejection. It motivates the equal-joint-budget validation in Section 6.

This three-gate arm has 19 H1 tasks at one noise level; the A-D arm has 18 H1 tasks at four noise levels. The S5 cache has no recorded failures for i.30.5 but lacks S2's separate replay-sentinel fields. Its within-arm counts are retained as exploratory evidence. Comparison of complete A-D and A-F panels requires common records.

Class-conditional interpretation. Within the S5 A/E/F arm at n=50, silence favors true over wrong forms by the likelihood ratio (2352/2600)/(1602/3800)=2.146. Under an explicitly hypothetical 50/50 class prior, the posterior probability of a wrong form is (1602/3800)/[(1602/3800)+(2352/2600)]=31.8%. The likelihood ratio measures evidence under this policy; the posterior also uses the assumed prior.

Table 15 additionally imposes $R^2 \geq 0.99$ and requires a finite far-field outcome. At n=50, 22,060/34,000 eligible H1 records pass that screen (64.9%). Gate-only silence and screened dangerous release therefore answer different questions with their own eligible populations.

5 Discussion

5.1 Match diagnostics to the mismatch direction

The controlled alternatives explain why a single method ranking is insufficient. A's mean-square column can cancel opposing curvature and has weak signal for odd-symmetric departures; interactions distinguish fitted-value augmentation from regressor-power augmentation (Table 4). Resampling calibrates a chosen statistic but does not expand the directions it examines. The task-level differences therefore matter alongside the average detection rates.

D is the strongest mean performer among A-D on the common S2 pool under the stated reference target. Its paired advantage over A is supported at n=200 and 1000. S5 separately shows that a richer quadratic rival has higher recorded mean detection than A in each displayed pool. Together these findings motivate testing both augmentation geometry and model-comparison alternatives. A paired D-versus-F ranking requires the common-policy experiment in Section 6.

5.2 Separate statistical assumptions from numerical stability

Native size is an empirical requirement: Tables 1-2 show departures from .05 even when the symbolic mean is a true monomial. The family-viability experiment supplies a second example. In rational coordinates, changing the resampling null substantially improves observed containment (Table 8). Transforming coordinates therefore requires reevaluating the noise and resampling assumptions.

Numerical implementation supplies a distinct failure mode. A stratified cross-check of exog-HC3 on 176 fits spanning all 13 H0 equations compared the original pinv(X′X) path, an SVD path, and statsmodels on

identical data. The SVD and statsmodels calculations agreed closely (maximum absolute p-value difference .0023), whereas the original path produced large rejection rates on near-collinear designs. Forming X′X amplifies conditioning problems; the stable path changes the empirical method comparison. The audit supports using the SVD results reported here. The complete measurements and earlier interpretation are preserved in Appendix G.

5.3 Measure useful rejection and residual release together

The diagnostics provide useful information. On the S2 evaluable pool, each gate's alarm precision for the specified far-field error exceeds the .2011 base rate (Table 7). After fit screening, 481/6,403 dangerous wrong models survive the four-gate policy. The same policy also releases many wrong forms with a small error at the chosen probe. These observations distinguish a structural label from a consequence measured at a particular input.

A useful workflow assessment reports both sides: dangerous models removed, dangerous models released, correct models retained, and declined evaluations. Table 15 connects these events on the same records. Table 16 supplies the complementary gate-only view. More gates can reduce silence while increasing false rejection; a common joint budget is needed to decide whether the added checks improve the trade-off.

Conditional rates retain their populations. The 7.5% screened-dangerous release rate measures failure to intercept dangerous candidates that already meet the fit screen, but it is a pooled figure driven by two tasks; excluding the threshold-borderline task bonus.15, the release fraction is 0% at n=200 and 1000. A task-level bootstrap over the 17 equations gives a 95% interval of [0%, 49.9%] for the pooled 7.5%, so the pooled figure is descriptive rather than precision- estimated. The 0.73% rate measures their frequency among all screened wrong forms. Deployment risk additionally depends on task prevalence and on the population of correct candidates, as the three-state accounting in Section 4.10 demonstrates.

5.4 Use family viability and observations as diagnostic follow-up

The battery returns a viable family set after coordinate-specific testing. On the recorded synthetic family truths, containment is approximately .93-.95; on out-of-family alternatives, Table 10 identifies structures that escalate and structures that retain a wrong family. This supplies a filtering stage for further model comparison, with overlap explicitly recorded.

The World Bank and engineering suites extend the software and diagnostic record to observed data. Their agreement and divergence patterns can guide follow-up measurements or alternative models. Without known form labels or controlled interventions, those patterns supply compatibility evidence. The strongest empirical performance conclusions remain those supported by the ground-truth suites.

6 Scope and validation priorities

6.1 Scope of the current evidence

The main comparison concerns a prespecified power-law family on positive data. S4 extends the diagnostic construction to four linearizable families at n=200; S5 examines one quadratic rival. The results describe these candidate families, the recorded sampling configurations, and their threshold policies. Residual permutation provides an empirical approximation under the evaluated conditions. More strongly dependent errors, additive-error models, or different rivals require their own calibration evidence.

S2 contains 18 eligible H1 tasks for detection and joint silence, and 17 with finite extrapolation outcomes for joint release. The excluded i.30.5 statistic-sentinel records and the missing i.26.2 far-field outcome remain in the audit trail. Repeated simulations estimate within-task behavior; task bootstrap intervals describe variation within the available equation pool. S3's 30 seeds per cell and S6's Monte Carlo repeats have correspondingly limited precision and scope.

The far-field outcome is a single probe at three times the observed training maximum of the first feature, with other log-features held at training medians. Consequently the evaluation coordinate can change across sample sizes and seeds. Threshold sensitivity and fixed-danger-label controls strengthen the descriptive result, but they do not turn this probe into an operating-domain error bound.

6.2 Two priority validation experiments

Independent evaluation of a common joint policy. The next comparison should compute the candidate gates on the same fitted records, choose thresholds on a separate H0 calibration split, freeze the policy, and evaluate on independent H0/H1 records. Strong single gates such as D and F should be compared with prespecified panels at the same target joint false-rejection budget. The final test should report actual false rejection, dangerous release, retained-candidate fraction, and declines, with paired task-level uncertainty. Holding out equations is needed when the intended claim concerns new equations. The existing S5 leave-one-equation checks provide limited transfer evidence to build on.

Evaluation on a fixed extrapolation domain. For each ground-truth equation, a physically valid external evaluation grid should be fixed before drawing the training samples. The same grid and prespecified error summary should be used across sample sizes, alongside the current endpoint probe. This would test the stability of the release pattern over an explicit domain. A paired training-sample design would further support analysis of the sample-size effect. Neither validation experiment is reported as completed in this manuscript.

6.3 Extension to discovered candidates and engineering decisions

A discovery-workflow claim requires evaluation of the expressions actually selected by a search algorithm. Candidate generation and selection should be frozen before independent diagnostic and prediction evaluation, and the diagnostic null should correspond to the candidate being assessed. The current main benchmark measures family diagnostics; algorithm-selected expression risk is a distinct next experiment.

The observational suites demonstrate data compatibility and diagnostic behavior. Engineering recommendations additionally require reference outcomes over the intended operating domain, relevant noise models, and a loss-based tolerance. These steps would connect the benchmark's measured failure channels to a specific application.

7 Reproducibility statement

The local benchmark source tree is D:/form_check. It contains generators, recorded configurations, raw and cached results, failure labels, and earlier analysis scripts. Appendix A reports the original fitted-run totals. S7 reuses S2/S5 records and adds no fitted datasets to those totals.

The manuscript bundle contains manuscript.md, build_pdf.py, audit_size.py, audit_results.py, and audit_s7.py. The audit scripts record source-file hashes, eligibility rules, thresholds, and counts in evidence/size.json, evidence/results.json, and evidence/s7.json. They reproduce the reported aggregation and Figures D.1-D.4 from the existing caches. Figure D.5 retains the historical timing measurements, with its inference-path scope stated. The build script renders the manuscript and figures.

Within the original source tree, runs/cd_svd/ contains the C/D SVD outputs; runs/srsd_full/ contains the S2 records and quality_recompute.parquet. S5 gate definitions are in gate_compare.py; joint_silence.py and joint_silence_toolbox.py identify the S7 arms. wb_run.py and wb_blindspot.py document the observational macro-data workflow; real_data/ contains engineering data manifests and source-specific license records.

The current validation includes record-level joins, input hashes, independent aggregation checks, selected implementation comparisons, and PDF consistency checks. Reproduction of the original fitting experiments is separate from re-aggregation of their stored outputs.

The artifact is publicly available at https://github.com/leokynlang/form_check. The data snapshot accompanying this manuscript is release tag data-2026-10-18. data_release/manifest.json lists every member path, size, and SHA-256, and SHA256SUMS.txt covers the archive. The repository README gives the clone, install, data-restoration, and aggregation commands, including audit_size.py, audit_results.py, audit_s7.py, reproduce_r2.py, verify_composition.py, threshold_decomposition.py, deep_analysis.py, and audit_task_intervals.py.

Third-party data are not uniformly redistributed: data_release/THIRD_PARTY_DATA.md and dataset_sources.json record per-source decisions, and copyrighted sources such as WebBook SRD 69 are source-download-only. Appendix G preserves the numerical and interpretive correction history.

8 Conclusion

This benchmark quantifies what selected specification gates detect, what false alarms they produce, and which dangerous wrong models survive the complete decision workflow. Controlled alternatives identify distinct blind spots. On the common 18-task physics pool, SVD exog-HC3 has the highest mean detection fraction among A-D at the stated H0-reference target, with positive paired D-A intervals at n=200 and 1000. Native-size and numerical checks show why calibration and stable inference must accompany that comparison.

The joint-release analysis measures the consequence of combining fit screening with valid diagnostic non-rejection. Of 6,403 screened dangerous wrong models, 481 pass all four gates: a 7.5% residual release rate under the specified endpoint criterion. The corresponding frequency among all screened wrong models is 0.73%. The released records are concentrated in two tasks; excluding the threshold-borderline task bonus.15, no dangerous model is released at n=200 or 1000, so the pooled 7.5% should not be read as a task-uniform failure rate. The gates remove many dangerous candidates, yet the remaining failure channel is observable and depends on the tested structure and decision policy.

The practical contribution is a framework for assessing diagnostic trade-offs on the same models, with failure registration and explicit conditional populations. Independent evaluation of common-budget panels and a fixed extrapolation domain are the next steps toward workflow recommendations. The benchmark provides concrete failure cases, comparative evidence, and a reproducible basis for those tests.

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Appendix A - Suite composition

Suite	Cells	Fits per cell	Total fits	Extra evaluations
S1 synthetic	285	500	142,500	-
S2 SRSD-Feynman	480	500	240,000	240,000 test sets; 232,500 extrapolation probes
S3 family-stratified	392	30	11,760	-
S4 battery	188	500	94,000	-
S5 gate comparison (v3)	102	200	20,400	-
S6 real data + probe (v3)	33	5	165	-
Total	**1,480**	-	**508,825**	**472,500**

Outside the seeded totals: the Appendix E engineering bench contributes 150 single-realization smoke cells (147 run), one fit per gate per cell.

Appendix B - Failure registration taxonomy

Reason	Meaning	Exercised by
`degenerate_zero_residual`	Zero-noise cell on exact ground truth; test undefined	S1/S2 $\sigma = 0$ cells
`rank_deficient`	Design matrix numerically rank-deficient	collinear designs
`insufficient_df`	$n - k < 1$ after augmentation	$n = 50$ multi-feature cells
`hc3_leverage_degenerate` / `hc3_singular_covariance`	HC3 undefined	degenerate built columns
`statistic_mismatch`	Wrapper statistic ≠ native statistic (drift sentinel)	continuous audit
`checked_false_*`	Native test declines to run (non-positive data, $n < 12$, non-finite)	not-applicable tasks
`runner_error` / generation failure	S5 SRSD-side generation exceptions	S5
indicator archived / no data	S6 indicator not published for a year	S6 (`EN.ATM.CO2E.KT` → AR5 replacement, logged)

All registered failures remain in the audit trail; denominators are documented per table. The battery's 94,000 and S5's 20,400 datasets contain zero data-layer failures.

Appendix C - Statistical uncertainty and eligible populations

Tables 1 and 2 contain the fit-level Wilson 95% intervals recomputed by audit_size.py, with 6,000 and 26,000 valid positive-noise H0 evaluations per n, respectively. The S1 zero-noise declines are excluded from size estimation. These intervals describe simulation precision on the fixed scenario/equation pools, not cross-equation generalization.

At S2 n=200, the equation-level analysis in audit_results.py resamples the 13 H0 equations 10,000 times, with 2,000 records per equation. For C, the mean is .053115 and the 95% task-bootstrap interval is [.050308, .056385]. For D, the mean is .056385 and the interval is [.054615, .058385]. Leave-one-equation-out ranges are [.052000, .053542] for C and [.055625, .056667] for D. Section 4.1 uses the same results with rounded presentation. A lower bound displayed as .050 is not proof of exact coverage of .05.

The paired D-A intervals in Section 4.2 instead resample 18 H1 equations, preserving each equation's paired difference. They must not be substituted for fit-level binomial intervals, or for results on a 19-task pool. Pointwise Figure D.3 intervals and paired differences address different quantities.

The joint-release ratios have a separate task-level analysis. audit_task_intervals.py resamples the 17 eligible H1 equations (B=10,000) and recomputes pooled Ld/S, Ld/Sd, and Ld/Da from each task's counts. With releases confined to two equations, the pooled Ld/Sd of 7.5% has a 95% interval of [0%, 49.9%], and 12% of resamples contain no release. This interval describes between- equation heterogeneity; it is not a binomial interval on 34,000 records and does not include threshold-estimation uncertainty.

Seeded computational repeats within one equation do not create additional independent laws. Macro and record-weighted averages coincide only where the task design is balanced. The old pinv-path uncertainty results are historical numerical-path diagnostics, not current method-performance intervals.

Appendix D - Result figures

Figures D.1-D.4 use the eligible record pools documented in audit_size.py and audit_results.py. Figure D.5 reports the historical timing measurements; its inference-path scope is specified in the caption.

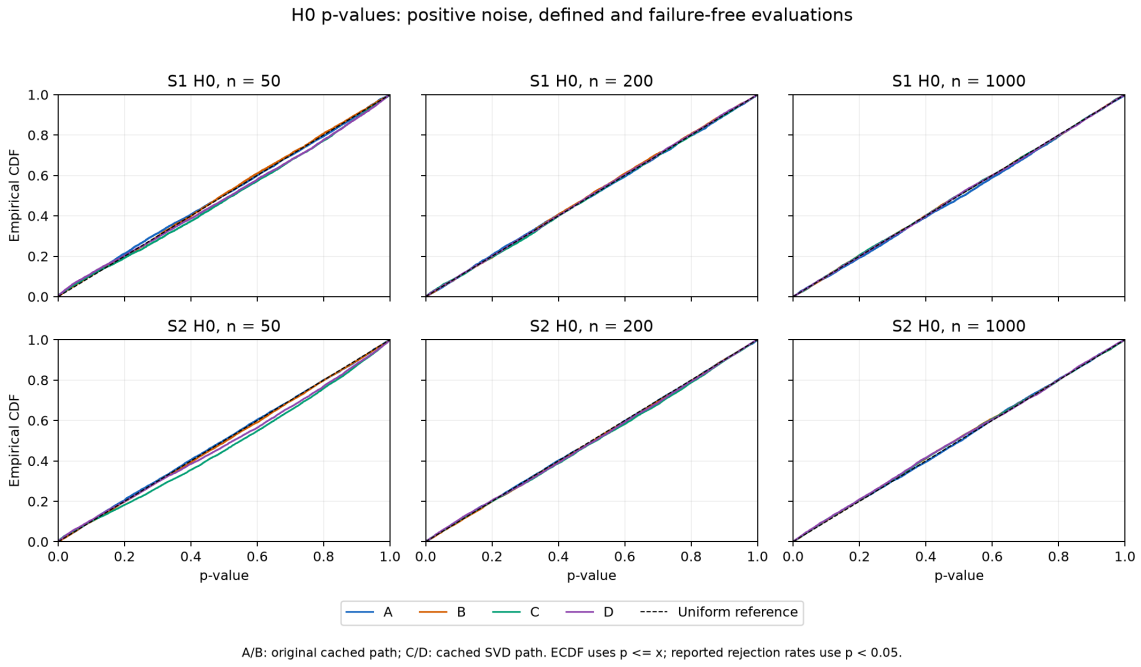


Figure D.1 - Empirical CDF of p-values under strict monomial H0 (uniform = calibrated): S1 synthetic H0 (left) and S2 SRSD-Feynman H0 (right). Approximate diagonal agreement does not establish exact 5% calibration; Table 2 reports the measured excess for C/D.

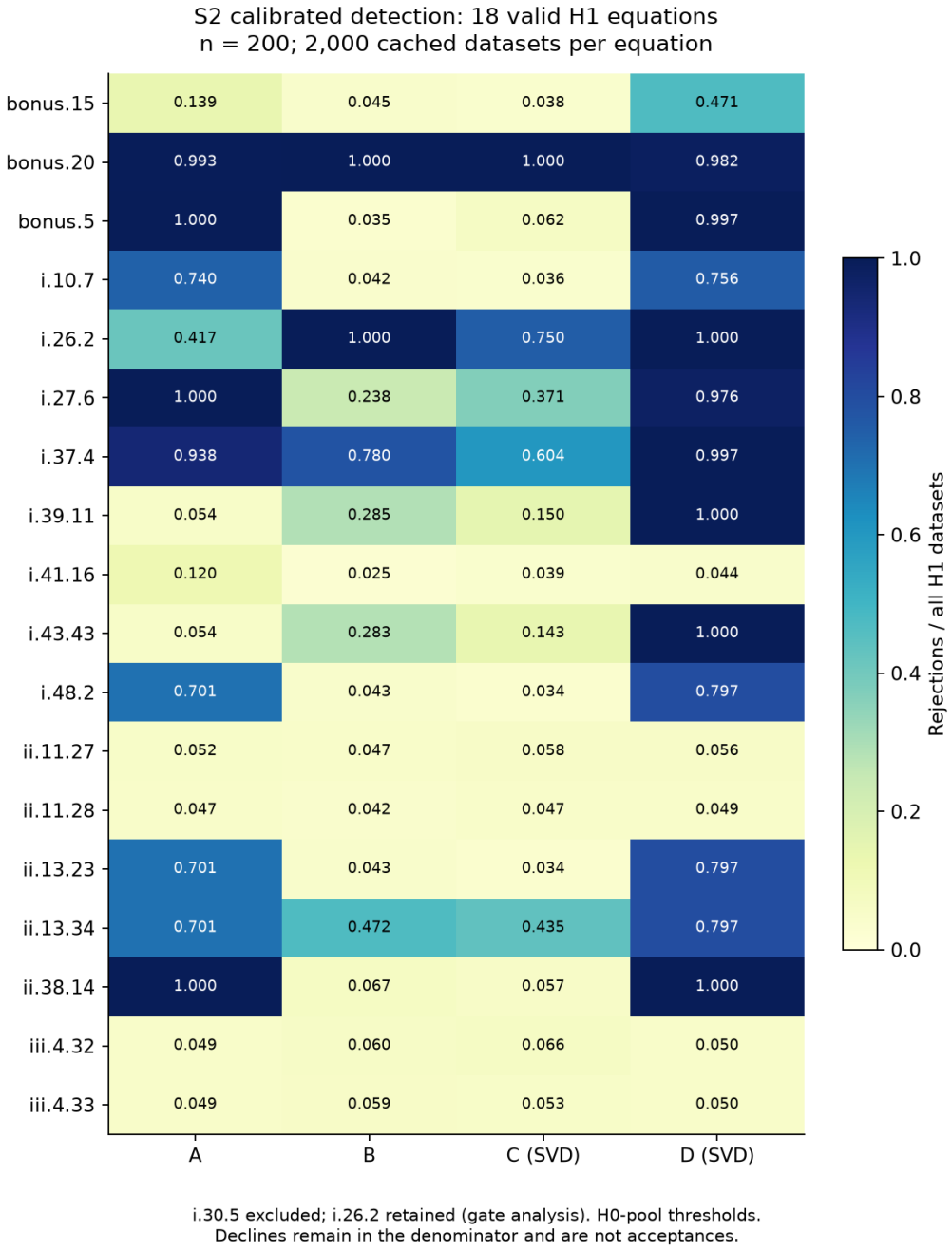


Figure D.2 - S2 H1 calibrated power per task (H0-pool 5% threshold, n = 200, rel > 0): one row per task, one column per gate; 18 sentinel-valid H1 tasks, excluding i.30.5. The two extrapolation-blind equations of §4.5 (i.39.11, i.43.43) are dark under A and B but reach 1.00 under D; iii.4.32 / iii.4.33 stay dark for every gate.

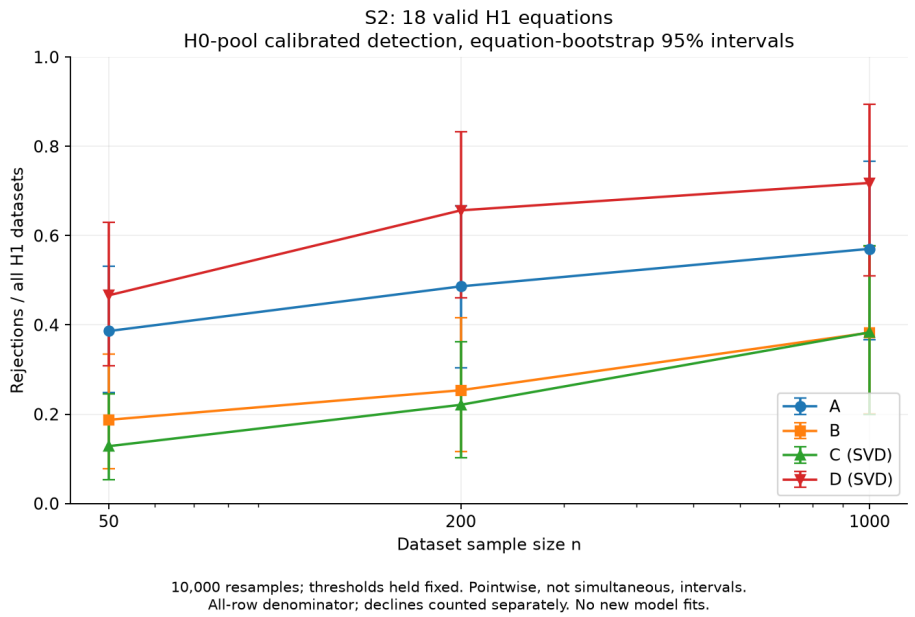


Figure D.3 - S2 H1 calibrated power versus n, with task-level bootstrap 95% intervals (10,000 replicates over the 18-task pool). Declines stay in the denominator but are not rejections. D has the largest mean in this pool; paired D-A intervals are in Section 4.2.

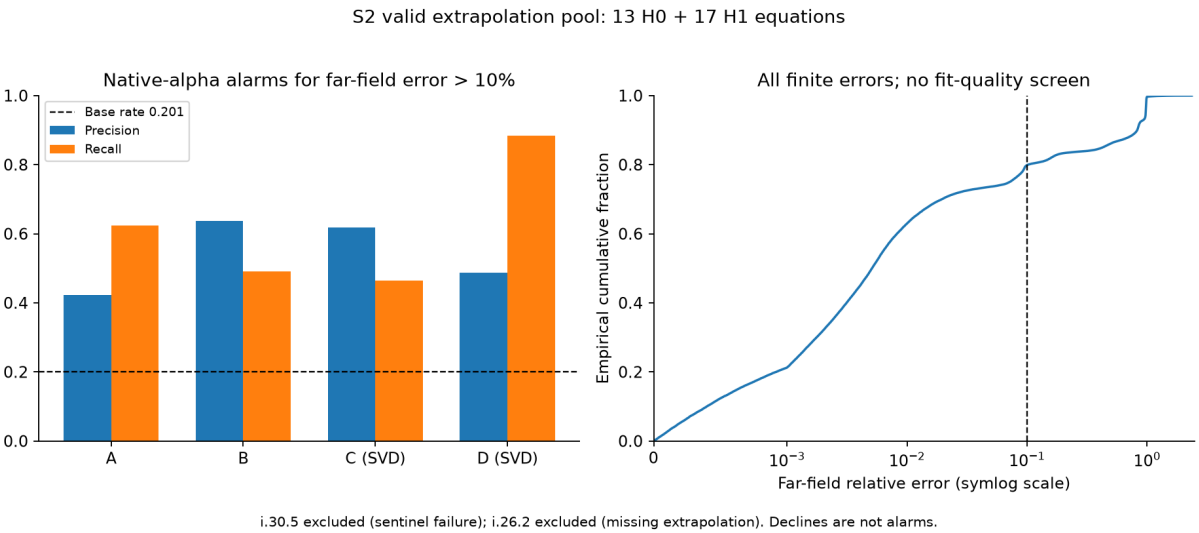


Figure D.4 - Left: alarm precision and recall for far-field error >10% on the revised S2 evaluable pool (13 H0 and 17 H1 tasks, positive noise, all n). Right: the corresponding error distribution. The task and missing-output filters match Section 4.10.

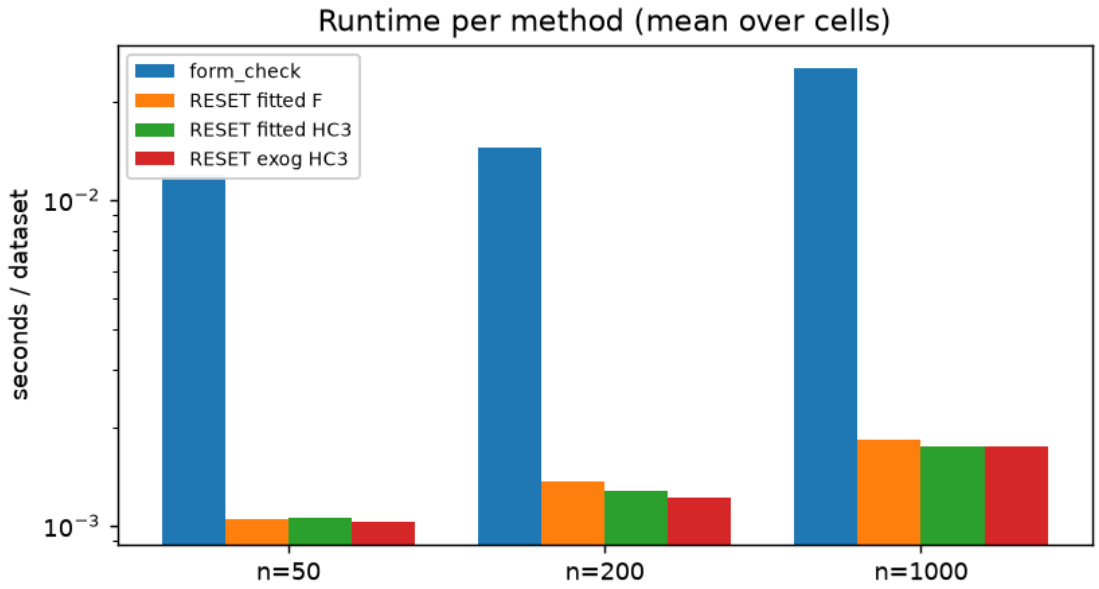


Figure D.5 - Per-dataset wall time; all four tests stay under 30 ms across n in the recorded historical S1 run. These timings do not establish equal runtime for the revised SVD inference path.

Appendix E - The real-engineering data bench (companion to S6)

S6 probes mechanism transfer on macroeconomic aggregates. The deployment domain these gates are actually built for is physical-engineering measurement - fluid, civil, hydraulic, energy, meteorological, seismic, chemical, and aerospace data, where power laws are the daily candidate models. This appendix documents a companion bench of that kind, run under the same contract as S6: real data has no ground truth, so the bench registers compatibility and divergence, not error rates; its cells are single realizations and stand outside Appendix A's seeded totals.

Composition and discipline. Nine open sources with source-specific licenses; 91 datasets; 150 candidate log-log relations registered in the local manifest with per-relation physical notes; 761,515 raw rows (Table E1). Raw bytes are cached (~115 MB) with SHA-256 hashes; cleaning is replayable and imputation-free - rows with missing or non-positive values are dropped and logged, and the few unit conversions are explicit in the manifest (e.g., Whiteside °C → K). Simulator datasets that violate the real-data discipline (a marine-engine and two wave-energy numerical models) are excluded and listed, as are WebBook fluid pages below the 10-row rejection-detection floor; one gauge whose instantaneous series was never published is logged as an empty skip. ERA5 cells are marked as reanalysis, not direct measurement.

Table E1 - Composition by source.

Source	Datasets	Relations	Rows	Examples
USGS NWIS rating curves	12	12	442,269	discharge ~ gage height, 12 gauges
UCI	11	38	145,920	CCPP, Tetouan power, steel, yacht, concrete
EPA fueleconomy	1	4	48,624	CO_2 ~ MPG; combined MPG ~ displacement
Open-Meteo / ERA5	10	20	87,600	100 m ~ 10 m wind speed (shear); gust factor
OpenML	3	8	21,997	aileron position; CPU utilization
USGS FDSN	1	3	12,334	felt reports, stations, intensity ~ magnitude
RDatasets	14	26	1,349	cars, trees, puromycin, stormer, rock
NIST StRD	13	13	968	certified nonlinear calibration curves
NIST WebBook	26	26	454	phase-transition enthalpies ~ temperature
Total	**91**	**150**	**761,515**	-

Protocol. Each cell applies the §3.2 gates unchanged at $\alpha = 0.05$ - A with its $B = 999$ permutation p, B/C/D as registered - on the positivity-filtered relation, subsampled to $n \leq 800$ (seed 42) to bound permutation cost. The run is a smoke pass: one realization per cell, compatibility plus first verdicts, not a rate estimate. Of 150 cells, 147 run; three are registered `insufficient_n` by design (Bennett5 and Roszman1 have no positive rows after the log-space positivity filter; Danwood has $n = 6$) - the bench's honest-failure probes working as intended.

Table E2 - Smoke verdicts on the 147 run cells.

Gate	Rejections	Registered failures
A - permutation curvature	97/147 (.660)	10 declines, all `checked_false_small_n` ($n \leq 11$)
B - RESET fitted F	111/147 (.755)	0
C - RESET fitted HC3	108/147 (.735)	0
D - RESET exog HC3	94/147 (.639)	3: `insufficient_df` ($n = 13$, four features), `hc3_leverage_degenerate` ($n = 546$), `rank_deficient`

A and B agree on 123/147 cells (92 both reject, 31 both quiet; 5 A-only, 19 B-only). The failure columns are themselves informative: D's three declines are exactly the three HC3 failure modes of Table 2 and Appendix B, now on observational data; A's ten are uniform declines-to-run on $n \leq 11$ cells; B and C never fail. Nothing fails silently - every declined evaluation is registered with a typed reason.

Designed regimes. The bench's design document assigns expected regimes to 66 of the 150 cells before the smoke pass (Table E3). The assignment is a reading aid, not a scored test: the H0-side laws are approximate by design, and at n up to 800 an approximate law is a detectable null. The 12 rating-curve cells - whose registered form $Q \sim h$ omits the documented stage offset h_0 of the true rating law $Q = C(h - h_0)^n$, a mismatch noted in the manifest - are rejected unanimously by A and B alike (12/12): detection of a documented misspecification, not a false alarm. The five H0-side cells A retains are those with no designed departure mechanism - the braking-distance v^2 law, the tree allometry, the road-load relation -

plus the two shear sites with the smallest fitted curvature.

Table E3 - Designed-regime cells, rejections at α = 0.05.

Designed regime	Cells	A	B	C	D
Approximate power laws (H0-side)	30	25	27	23	23
Known non-power-law physics (H1-side)	36	20	24	24	14

(A's H1-side count includes its nine WebBook declines at n ≤ 11 as non-rejections.) On the H1 side no gate sweeps the board: the misses concentrate in WebBook enthalpy cells whose measured temperature window keeps the local power-law approximation good - e.g., toluene Δ_vapH ~ T: all four gates quiet at n = 26 - a mutual-miss cell recorded as measured.

Table E4 - Named cells: mechanism echoes and divergences; p-values at α = 0.05 in A / B / C / D order, - = declined with registered reason.

Cell	A / B / C / D	Reading
Wind shear $v_{100} \sim v_{10}$, 10 ERA5 sites (n = 800 ea.)	8/10 / 10/10 / 7/10 / 7/10	B flags every site ($p \leq 2\times10^{-4}$); A retains exactly the two smallest-curvature sites (.23, .41) - a graded response, not a threshold artifact
Gust factor ~ v_{10}, same sites (n = 800 ea.)	10/10 ×4	unanimous rejection at every site; no truth to adjudicate
puromycin rate ~ conc (Michaelis-Menten), n = 23	**.011** / .069 / .169 / .169	A detects the saturation curvature, B misses at α = .05 - Table 3's small-sample power gap, live
stormer Time ~ Viscosity + Wt (calibration law), n = 23	.001 / 7×10^{-4} / 9×10^{-3} / -	single-feature cells quiet (A .94, B .99); the second feature carries the interaction the law contains
EPA CO_2 ~ MPG (stoichiometric inverse, exponent −1), n = 800	.030 / 1.5×10^{-4} / .727 / .727	A and B reject on a near-exact physical law bent by fleet drive-mix curvature
EPA combined MPG ~ displacement (road load), n = 800	.386 / .596 / .599 / .599	designed power law passes
CCPP PE ~ V, n = 800	.642 / 8×10^{-4} / ≈0 / .023	proxy curvature: V correlates .81 with the strongly curved AT, so B/C's fitted-power columns inherit AT's curvature while A's V-only direction set cannot - the mirror of dilution
automobile horsepower ~ engine size, n = 203	.878 / 1.2×10^{-4} / .094 / .094	divergence under heteroskedastic body-style scatter
rock perm ~ area + peri, n = 48	.835 / 1×10^{-5} / 5.6×10^{-6} / .596	A/D quiet, B/C reject; no truth to adjudicate
housing price ~ 4 features, n = 546	.723 / .683 / ≈0 / -	D's leverage-degeneracy failure mode, live at n = 546, while A/B/C return verdicts

Scope. What the bench supports is what S6 supports: compatibility at scale (147/150 cells run across nine sources, zero silent failures), failure registration on observational data, and mechanism-echo cells whose directions match the synthetic map. Its per-gate rejection totals (97/111/108/94 of 147) are compatibility statistics on single realizations without ground truth - they rank nothing, and reading them as performance would repeat exactly the leaderboard confusion the benchmark exists to correct - A's observed non-rejections (the shear sites, the road-load law, puromycin at n = 23), A's interaction and dilution behavior (stormer, CCPP), D's structural fragility (all three failure modes live). What it cannot support: rates, calibration, or rankings - single realizations, no ground truth, and time-ordered NWIS/Open-Meteo series whose permutation-independence assumption is an approximation marked in the manifest. All four p-values per cell are recorded locally (`real_data/smoke_results.csv`) with the manifest (`real_data/clean/manifest.json`, SHA-256 per source) and replayable fetch/prepare/smoke scripts.

Appendix F - Core code listings

Abridged reference implementations (error handling and instrumentation omitted; full implementations are present in the local benchmark project). Listing 1 is the wrapper around the project's native `form_check`: it recomputes the curvature statistic in log-log space, verifies it against the native value (drift sentinel), and derives the empirical p-value from a vectorized residual-permutation null. Listing 2 is one battery member: the same statistic in a family's linearizing coordinates with a wild bootstrap null.

Listing 1 - Method A wrapper: permutation curvature test (common.py, abridged).

```
def form_check_with_p(data, target, features, seed, b_perm=999):   # (abridged)
    # Method A wrapper: statistic = coefficient of the mean-square column
    # in log-log space; null = residual permutation, vectorized via pinv row.
    fc = dimreflex_form_check(data, target=target, features=list(features),
                              n_boot=200, random_state=seed)
    if not fc.get("checked", False):
        return {..., "fail": classify_reason(fc["reason"])}

    lx = np.log(data[list(features)].values)
    ly = np.log(data[target].values)
    n = len(ly)
    quad = (lx[:, 0] - lx[:, 0].mean())**2 if len(features) == 1 \
           else ((lx - lx.mean(axis=0))**2).mean(axis=1)
    base = np.column_stack([np.ones(n), lx])
    design = np.column_stack([base, quad])

    coef, *_ = _ols(design, ly)
    c2 = float(coef[-1])                    # statistic: curvature c2
    if not np.isclose(c2, fc["curvature"], rtol=1e-6, atol=1e-10):
        return {..., "fail": "statistic_mismatch"}   # native-drift sentinel

    bcoef, *_ = _ols(base, ly)              # restricted fit -> perm null
    pred, resid = base @ bcoef, ly - base @ bcoef
    if float(resid @ resid) <= 1e-12 * float(ly @ ly):
        return {..., "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(int(seed) * 1_000_003 + 17)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in y
    c2_base = float(pinv_row @ pred)
    idx = np.argsort(rng.random((b_perm, n)), axis=1)
    null_c2 = c2_base + resid[idx] @ pinv_row    # B permutations at once
    mu, sd = float(null_c2.mean()), float(null_c2.std())
    p_emp = (1.0 + float(np.sum(np.abs(null_c2 - mu)
                                >= abs(c2 - mu)))) / (b_perm + 1.0)
    return {"p": float(p_emp), "rej": bool(p_emp < ALPHA),
            "t": float((c2 - mu) / sd)}
```

Listing 2 - Battery member: family-coordinate curvature test with wild bootstrap null (battery.py, abridged).

```
FAMILIES = {                                    # (u(x), v(y)) per family
    "power":       (np.log,          np.log),
    "exponential": (lambda x: x,     np.log),
    "logarithmic": (np.log,          lambda y: y),
    "rational":    (lambda x: 1.0/x, lambda y: 1.0/y),
}

def spec_test_wild(X, y, uf, vf, seed, b_wild=999):   # one battery member
    # H0: conditional mean linear in family coordinates (u, v).
    # Statistic: coefficient of quad = mean_j(U_j^2); null: wild bootstrap.
    U = _standardize_cols(uf(X))         # per-column transform + standardize
    v = vf(y)
    if (X <= 0).any() or (y <= 0).any():
        return {"p": np.nan, "fail": "nonpositive"}
    base = np.column_stack([np.ones(len(v)), U])
    quad = (U**2).mean(axis=1)
    design = np.column_stack([base, quad])
    coef, *_ = np.linalg.lstsq(design, v, rcond=None)
    t_obs = float(coef[-1])
    b, *_ = np.linalg.lstsq(base, v, rcond=None)
    resid = v - base @ b
    if float(resid @ resid) <= 1e-12 * float(v @ v):
        return {"p": np.nan, "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(seed)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in v
    c2_base = float(pinv_row @ (base @ b))
    signs = rng.choice((-1.0, 1.0), size=(b_wild, len(v)))
    null = c2_base + (resid[None, :] * signs) @ pinv_row   # Rademacher wild
    mu, sd = float(null.mean()), float(null.std())
    p = (1.0 + float(np.sum(np.abs(null - mu) >= abs(t_obs - mu)))
         ) / (b_wild + 1.0)
    return {"p": float(p), "t": float((t_obs - mu) / sd)}
```

Appendix G - Audit and correction history

G.1 Statistical, provenance, and presentation corrections

This ledger supersedes the mixed v2-v5 change notes in the supplied working draft. The original PDF and Markdown are preserved unchanged. Historical scripts and reports remain evidence of how earlier summaries were produced, not automatic authorities for the current numbers.

1. **Task pools and failures.** Record-level joins identify 7,500 S2 statistic-sentinel failures for feynman-i.30.5 over the full grid. Revised power and joint silence use 18 H1 tasks; joint release additionally requires a finite far-field outcome, leaving 17 H1 tasks because i.26.2 lacks that outcome. Original generated-fit totals are not relabeled as eligible observations.

2. **Joint-release accounting.** Table 15 now defines all four counts and three conditional rates on the same eligible records. Its pooled counts are A=65,992, B=6,403, C=481, D=36,169 (shown as S, Sd, Ld, Da in R2). Gate-only silence is not substituted for complete-pipeline release. Comparisons between different conditional fractions no longer support a claim of danger-concentrated failure.

3. **S7 provenance and gate sets.** S7 is a reanalysis of existing S2/S5 records, not new sampling. The S5 arm uses A/E/F, not A-F. A paired A versus A/E/F comparison replaces the invalid cross-arm interpretation. Likelihood ratios and equal-prior posteriors have separate formulas. The S5 cache's weaker

historical sentinel provenance is disclosed.

4. **SVD power and calibration.** Table 3 and the paired bootstrap use one 18-task pool and explicit decline handling. D-A effects are recomputed from the SVD data, not obtained by reversing old signs. The n=200 H0 task-level intervals are consistent between Section 4.1 and Appendix C. The manuscript no longer describes all C/D Wilson intervals as covering .05. Table 1 is also corrected: the 1,500 zero-noise declines per n are excluded, leaving 6,000 evaluated H0 records, rather than diluting the rejection rate with undefined tests.

5. **Claims, references, and artifacts.** The introduction and related work distinguish existing structural-recovery evaluation from this paper's diagnostic-gate experiment. References [6] and [11] have corrected authorship. Sample-size effects are not attributed solely to screening. Local script presence is not described as a public archive or a completed independent replication. The original scripts are not modified.

6. **What was checked in this revision.** The accompanying audit_size.py, audit_results.py, and audit_s7.py re-aggregate existing caches, record input hashes and task filters, and generate machine-readable evidence. build_pdf.py renders the corrected manuscript and figures. No new model fitting, public release, or engineering safety validation was performed. Remaining research limitations are in Section 6.

7. **Research presentation revision (R2).** R2 reorganizes the manuscript around detection, false-alarm cost, and extrapolation-related joint release. The abstract and introduction state the three experimental contributions; the discussion presents method-selection lessons; Section 6 specifies two priority validation experiments and the conditional extension to discovery-generated candidates. Historical accounting corrections are retained in this appendix. All numerical data rows in Tables 1-16 and the companion engineering tables are preserved from R1. This revision adds no model fits or completed validation experiments. Appendix H reports an exploratory PySR counterpart outside the main tables.

8. **Review-response revision (R2, 2026-10-18).** The artifact is public at https://github.com/leokynlang/form_check with data release tag data-2026-10-18. Section 7 now points to it; reference [17]'s licensing description is corrected; Section 4.10 adds a pooled task-level threshold sweep; the abstract, introduction, discussion, and conclusion state the task concentration of the 481 releases; and Appendix H adds the exploratory PySR counterpart. No aggregate (S, Sd, Ld, Da), table row, or figure in Tables 1-16 changed.

G.2 Numerical implementation audit

The implementation audit compares the original pinv(X′X) calculation, an SVD calculation using pinv(X), and statsmodels HC3-Wald with the same F reference. The exog-HC3 check spans all 13 H0 equations, stratified by equation, sample size, and noise level (176 fits on identical input data). Maximum absolute p-value difference between the SVD and statsmodels paths is .0023; maximum relative statistic difference is .0038; covariance Frobenius relative difference is at most 3×10^{-4}. These measurements establish close agreement on the tested fits.

The original path rejects at rate 1.00 on seven tasks whose measured cond(X′X) is at least 4×10^{17}, with measured cond(X) from 6.5×10^{8} to 3.2×10^{10}. On these small audit samples, both stable paths have task

rejection rates from 0 to .10. The two better-conditioned tasks ii.27.16 and ii.27.18, with cond(X) near 10, show no corresponding failure. A fitted-HC3 check on two tasks separately gives 8/120 rejections (.067, Wilson [.034, .126]).

Forming X′X squares the spectral condition number before the pseudoinverse cutoff is applied. On the observed near-collinear designs, discarded small singular directions alter the hat and sandwich calculations. The audit attributes the large original-path distortion to this numerical realization; it does not establish a failure of the HC3 correction. The SVD path supplies the C/D method comparisons reported in the main tables.

The earlier S2 H0 summaries for the original path were C=.417-.422 and D=.563-.565 across sample sizes. These historical values are withdrawn as evidence about HC3's statistical performance. The implementation audit was a separate, one-off comparison rather than a rolling software check on every fit. Close implementation agreement and empirical size are different validation questions.

Appendix H - Exploratory PySR counterpart

This appendix reports an exploratory check that applies the four gates to expressions emitted by a symbolic-regression search. It is separate from the main benchmark: candidate generation is not frozen, the sample is small, and the counts are compatibility evidence rather than rate estimates. The data are in evidence/deep_analysis.json and runs/pysr_gate/wide_svd/ in the data bundle.

Protocol. PySR searched the log-space training design with a restricted arithmetic operator set (+, -) matching the power-law hypothesis class. Each emitted expression was refitted as a prespecified candidate and passed through gates A-D under the same native thresholds. The run produced 360 candidates across the evaluated tasks.

Results. All 360 candidates: 87 jointly silent (24.2%). In the $R^2 \geq 0.99$ layer (183 candidates), 70 are jointly silent (38.3%), close in magnitude to the 38.6% S2 four-gate figure at n=50 although the candidate pools differ. In the far-field error >10% layer (142 candidates), only 3 are jointly silent (2.1%); the $R^2 \geq 0.99$ and far-field >10% intersection contains 2 candidates, both rejected by gate D ($p \approx 5\times10^{-29}$) while A, B, and C are silent. Per-gate rejection over all candidates: A 182 (50.6%), B 103 (28.6%), C 82 (22.8%), D 189 (52.5%); A is the sole rejecting gate on 45 candidates and D on 41.

Scope. These counts describe one search configuration and 360 candidates, of which only two meet the joint danger criterion. They do not establish a pass rate for discovery algorithms, do not rank the gates, and are not merged into Tables 1-16. Their purpose is to show that the same diagnostic record can be attached to search-selected expressions and that the main benchmark's failure channel has a counterpart in this pilot.